Stage 3 Data Consistency & QA Validation Report

Data Engineer role — preparing and checking the data

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Project: Customer Churn Analysis for Advanced Telecommunications Solutions (ATS)

1. What this report is about

My role on this project is Data Engineer. In Stage 2, I cleaned and prepared the customer data and handed it over to the Data Analyst, who used it to build and refine the churn prediction model in Stage 3.

My job is to make sure the data the model was trained on is correct, consistent, and free of problems. If the data going in has issues, the model's results cannot be trusted, no matter how well it was built.

This report covers seven checks I ran on the handed-over data, comparing it against the model-builder's own notes and my original Stage 2 documentation, to confirm everything lines up as expected.

2. Summary of what was checked

Check	What we found	Result
Row counts match what was documented	6,741 total → 5,392 training / 1,349 testing	PASS
Churn rate stayed balanced across the split	26.57% full / 26.58% training / 26.54% testing	PASS
No customer appears in both sets	0 overlapping rows	PASS
Scaled and unscaled files match up correctly	Same rows, same outcomes in both versions	PASS
Number scaling was done the right way	Recalculated it independently — matches exactly	PASS
No missing information	0 blank values across all 5 files	PASS
No duplicate customer records	0 exact duplicates found	PASS
The columns the model uses actually exist and are named correctly	All three feature groups line up with what Stage 2 delivered	PASS

Overall result: every check passed. No data quality problems were found, and the model was built on a clean, correctly prepared dataset.

3. The checks explained in more detail

3.1 Row counts and how the data was split

I reloaded all five files handed over from Stage 2 and compared the number of rows against what was documented, and against the model-builder's own counts inside their notebook.

- Full prepared dataset: 6,741 customer records, 26 columns, matching the Stage 2 report.
- Split into training and testing sets: 5,392 and 1,349 records — exactly an 80% / 20% split, as intended.
- The scaled and unscaled versions of each set contain the same customers in the same order, just with the numbers transformed differently.

3.2 Keeping the churn rate balanced

Churn rate: 26.57% in the full dataset, 26.58% in training, 26.54% in testing — all within a tenth of a percentage point of each other. This confirms the data was split carefully so that both the training and testing sets have a fair, representative mix of churned and retained customers. If this hadn't been done properly, the model's results could have looked better or worse than they really are, purely because of how the data happened to be split.

3.3 Checking for any overlap between training and testing data

I compared every row in the training set against every row in the testing set and found zero matches. This means no customer record was used both to train the model and to test it — an important condition for trusting the test results as a genuine measure of how the model would perform on new customers it hasn't seen before.

3.4 Confirming the number scaling was done correctly

Some columns (like tenure and monthly charges) needed to be rescaled so the model could work with them properly. The correct way to do this is to work out the scaling using only the training data, then apply that same scaling to the testing data — never the other way around.

- I independently redid this scaling from scratch, using only the unscaled training data, and compared the result to the files that were actually delivered. They matched exactly, confirming the scaling was done the right way with no shortcuts.
- As expected, the training data's scaled values centre almost exactly around zero, since the scaling was based on that data.
- The testing data's scaled values are close to, but not exactly, centred around zero — which is exactly what should happen when a scaling method learned from the training data is applied to a different set. This is the expected sign that no test information was used to influence the scaling.

3.5 Missing information and duplicate records

Zero missing values and zero duplicate rows were found across all five files. This matches what was already established during Stage 1 and Stage 2, meaning nothing was lost, corrupted, or accidentally duplicated during the hand-off to Stage 3.

3.6 Making sure the right columns are being used

The model-builder's notebook works with three different groupings of columns for different experiments. I checked each grouping against what Stage 2 actually delivered and against my earlier statistical testing:

- A basic set of 12 columns — all present and correctly named.
- The full set of 20 columns from Stage 2 (everything except the outcome being predicted) — nothing was missing or silently left out.
- A reduced set of 17 columns, which leaves out three columns (gender, phone service, and multiple lines) because earlier testing showed these had no meaningful relationship with churn. This matches my own findings from Stage 2 exactly.

No mismatches were found between what the model-builder assumed about the data and what was actually delivered.

4. Data quality issues found

None. All seven checks passed. No fixing, re-labelling, re-scaling, or removal of rows was needed at this stage.

5. Points worth keeping in mind (carried over from Stage 2)

Two things flagged back in Stage 2 are still worth keeping in mind when reading the Stage 3 results and writing the final report:

- Contract type and internet service type did not show a statistically meaningful link to churn in this particular dataset, even though they usually do in telecom data generally. This is worth remembering when discussing which factors matter most for churn, since it goes against what is typically expected.
- Contract length (in months) and total number of services were kept through Stage 2 to support customer segmentation, even though on their own they showed very little connection to churn (statistical significance scores of 0.068 and 0.900). If these appear at all important in the model's results, that should be read as helping with segmentation rather than being strong churn indicators on their own.

6. Conclusion

The data handed over from Stage 2 — all five files, along with the scaling and preparation documentation — is fully consistent with how it has been used in the Data Analyst's Stage 3 model-building notebook. Row counts, the balance of churned customers, the absence of any overlap between training and testing data, correct number scaling, and the columns used all check out against the original Stage 2 documentation and code.

No further action is needed. This hand-off is confirmed as reliable and ready to support the Stage 3 predictive modelling deliverable and the team's final report.